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Machine Learning Optimizations to Damped Energy Terms in the Effective Fragment Potential Method

Introduction and Background

- Calculating non-bonded molecular interactions is expensive
- SAPT provides benchmark accuracy but comes at high computational cost
- Effective Fragment Potential (EFP) method¹ offers a cheaper alternative using damped terms
- Goal: Apply ML to improve damping functions, proof-of-concept with overlap-based dispersion**

$$E_n^{EFP} = [E^{Coul} + E^{Pol} + E^{Disp}] + E^{ExRep} + E^{CT}$$

These three have damping functions

Dispersion energy (E^{Disp}) has three terms

$$E^{Disp} = E(R^{-6}) + E(R^{-7}) + E(R^{-8}) = \sum_{n=6,7,8} E(R^{-n})$$

Each term is of the form
damping function * analytic expression

$$E(R^{-n}) = f_n^{disp} * analytic\ expression$$

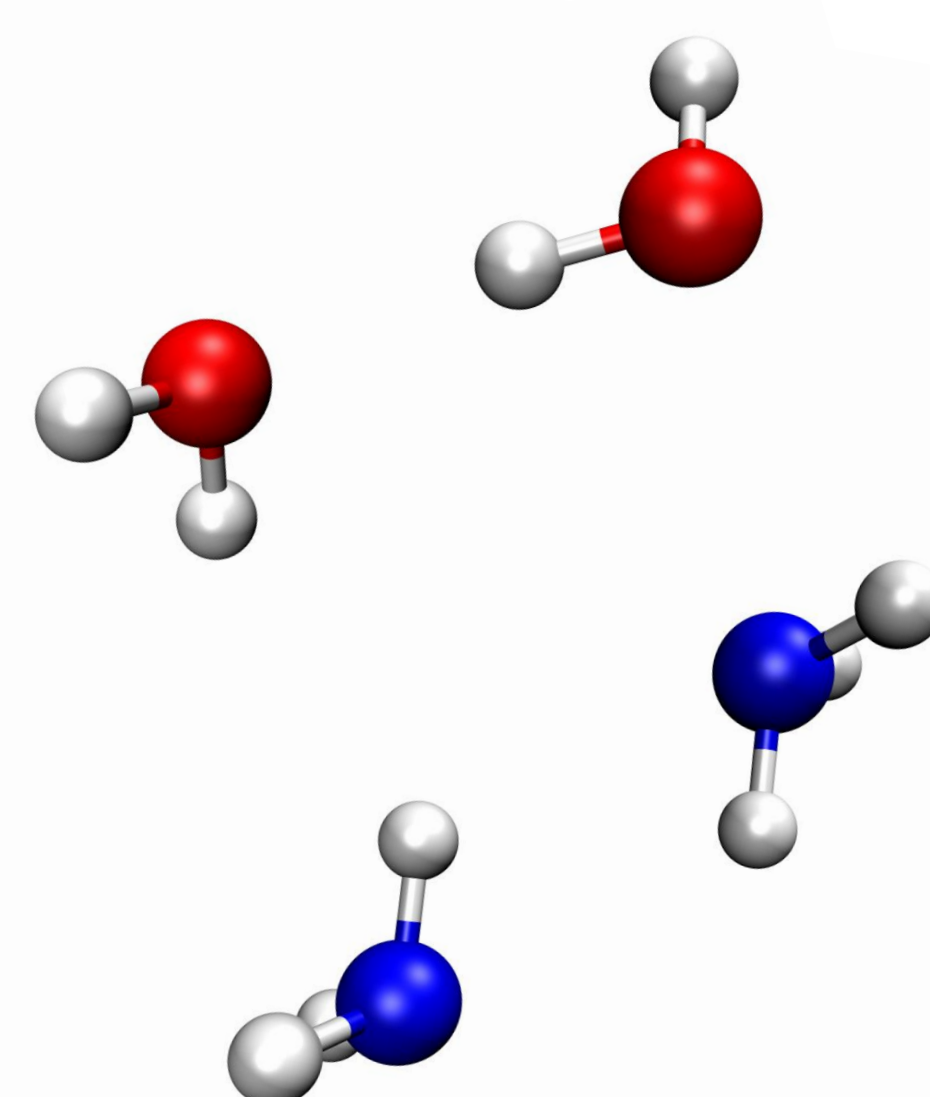
Two types of f_n^{disp} :

Overlap-based

$$f_n^S(i, j) = 1 - S_{ij}^2 \sum_{n=0}^{6\ or\ 7\ or\ 8} \frac{-2\ln|S_{ij}|^{n/2}}{n!}$$

Tang-Toennies

$$f_n^{TT}(i, j) = 1 - \left(\sum_{n=0}^{6\ or\ 7\ or\ 8} \frac{(bR_{ij})^n}{n!} \right) e^{-bR_{ij}}$$



Methods

Feature Engineering

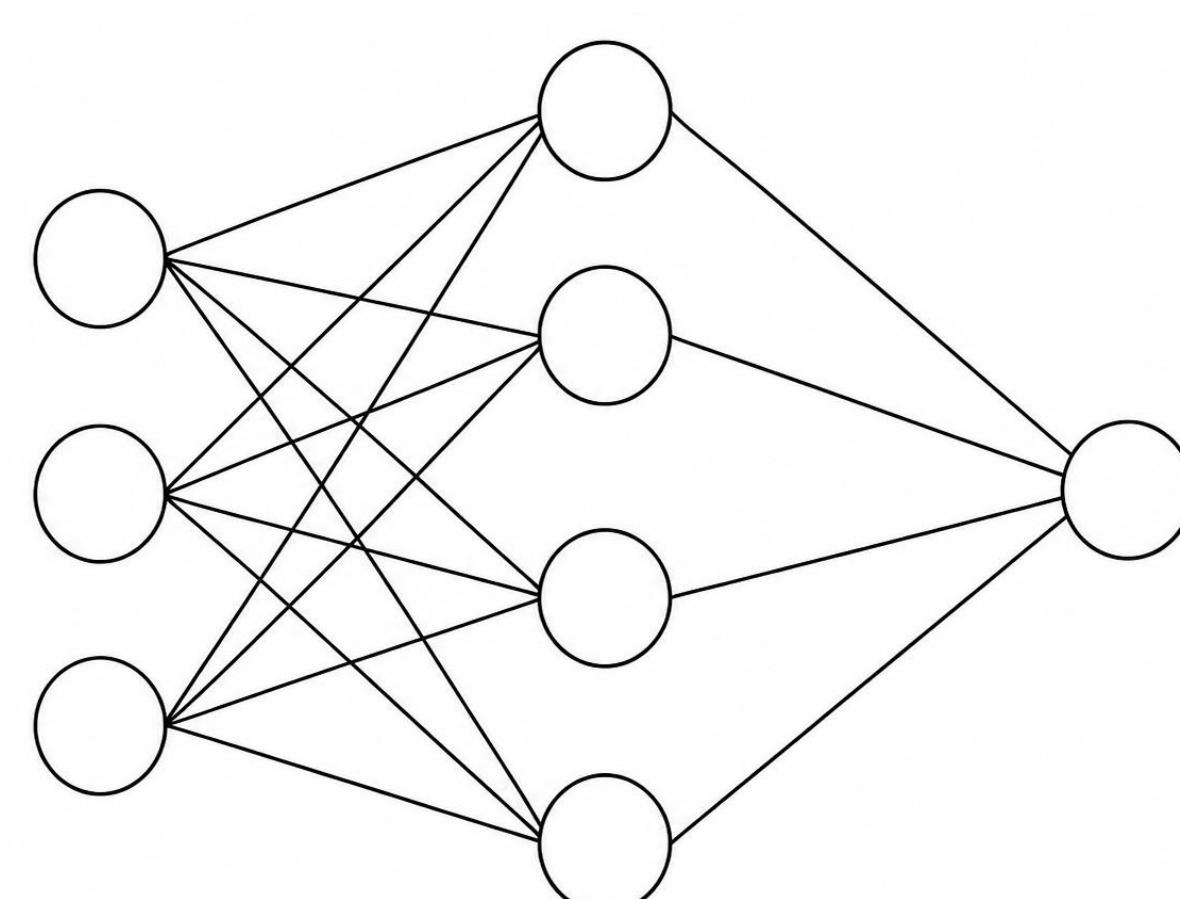
- S_{ij} : Two-body term for each LMO pairing in an interaction
- $\sum |S_{ij}| * (E(R^{-6}) + E(R^{-7}) + E(R^{-8}))$
- $\sum S_{ij}^2 * (E(R^{-6}) + E(R^{-7}) + E(R^{-8}))$
- $\sum \ln|S_{ij}| * (E(R^{-6}) + E(R^{-7}) + E(R^{-8}))$

Model

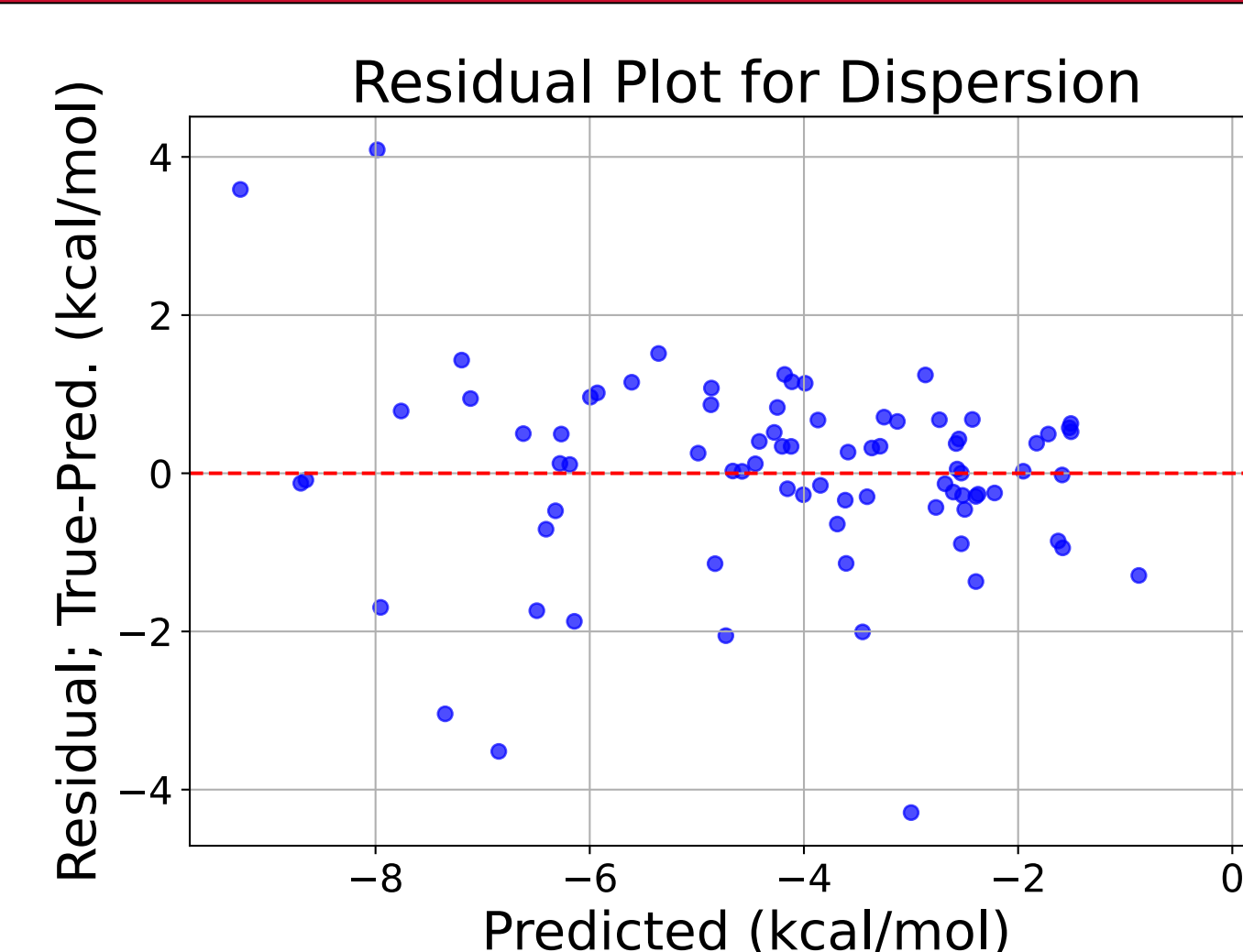
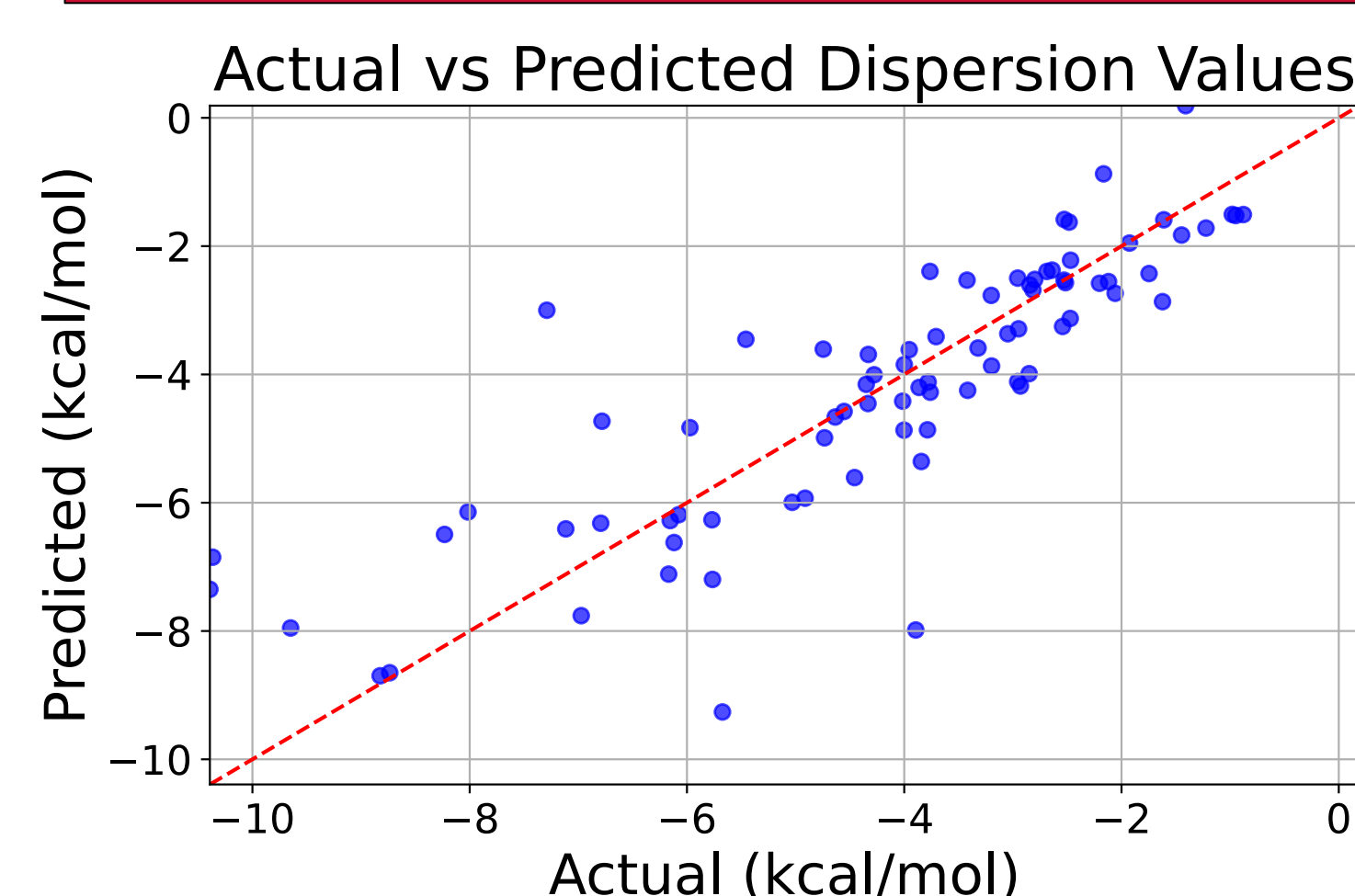
- StandardScaler
- MLPRegressor neural network
- Architecture:** One hidden layer, four neurons
 - ReLU activation
 - LBFGS solver
 - $\alpha = 0.01$

Validation

- LOOCV:** Leave-one-out cross validation
 - Implemented during training to maximize data usage & eliminate bias
- Performance metrics

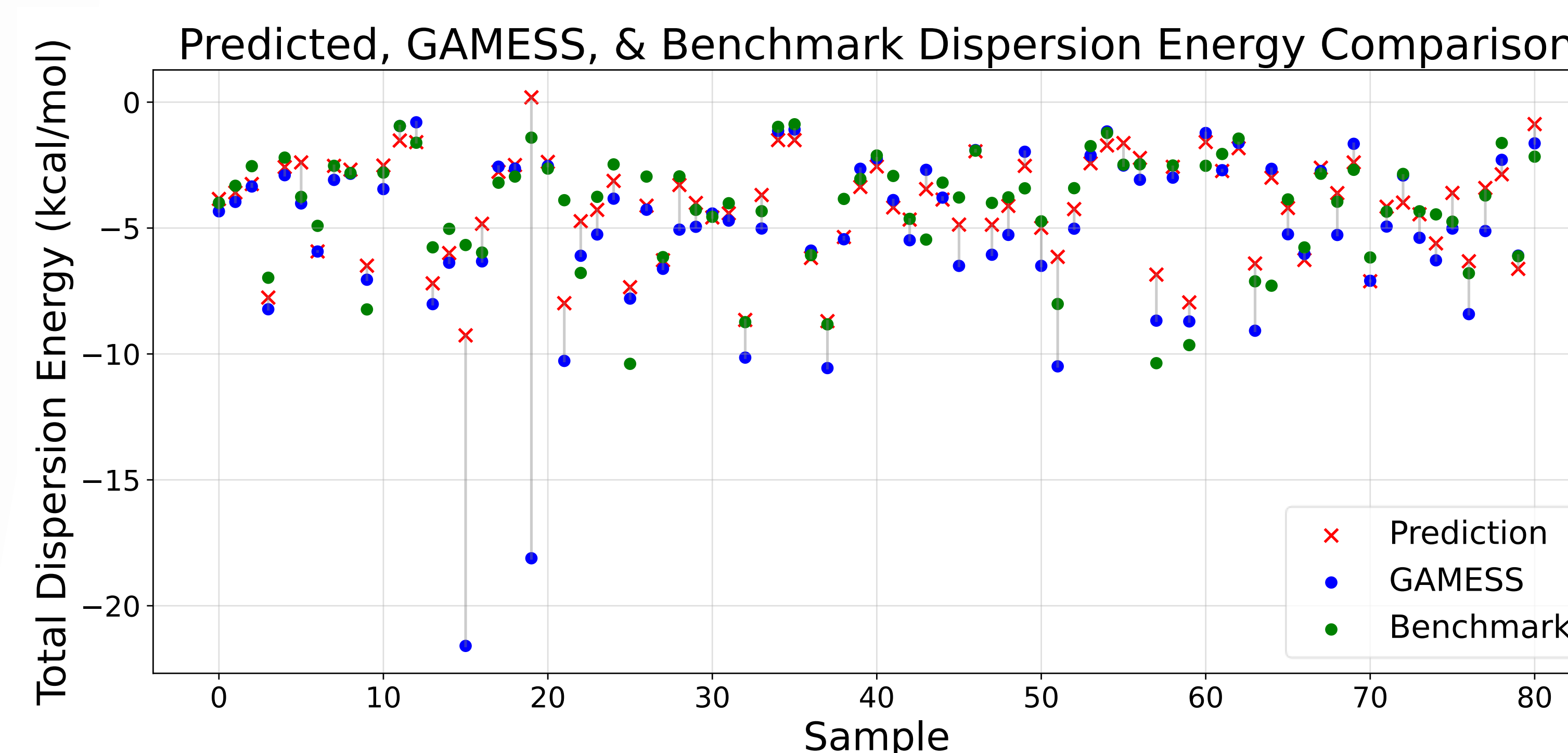


Results



- Strong agreement with benchmark values, few major outliers
- Residuals centered near zero, randomly scattered w/ no noticeable trend
- Reduction in R^2 score from training to LOOCV suggests underfitting
- Lower RMSE, MAE imply good accuracy

Metric	Value
Training R^2	0.842
Leave-One-Out CV R^2	0.683
RMSE	1.236
MAE	0.855



Conclusions and Future Directions

Conclusions

- MLP damping predictions closely matched benchmark dispersion energies, outperforming GAMESS damping in select cases
- Strong outliers remain, model is imperfect
- Successful proof-of-concept for applying ML to the EFP method

What I Learned

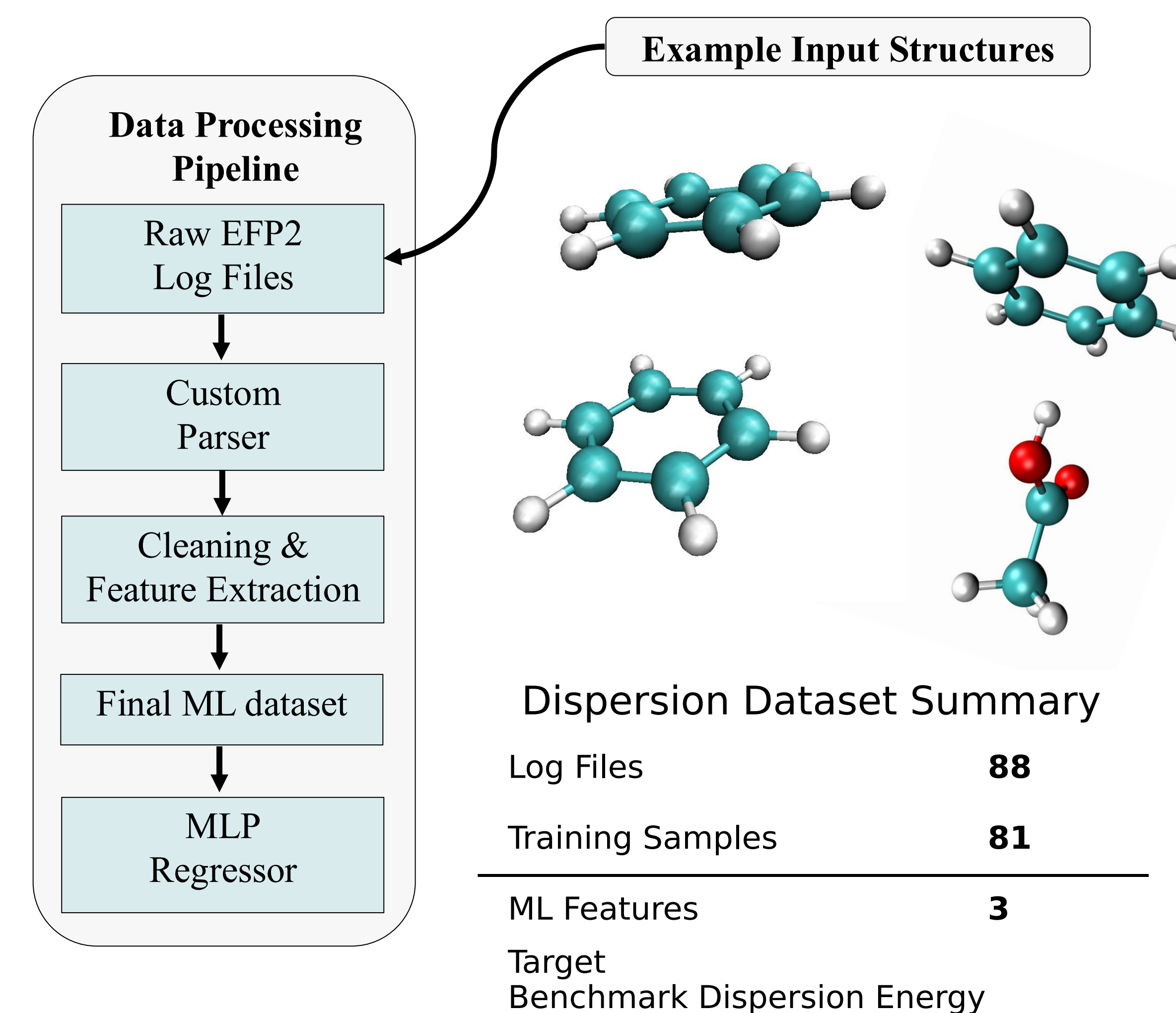
- Chemistry required for success (dimers, interaction energy, bonds, etc.)
- Implementation of LOOCV to increase accuracy and properly validate data
- Collaboration and teamwork in a research setting

Future Directions

- Increase dataset size
 - Incorporate remaining interactions
 - Add more benchmark sets like the A24
- Extend to remaining damped energy terms in EFP method
- Compare against alternative regression models

Dataset and Pipeline

- Wrote custom parser for GAMESS EFP2 log files
- Dataset from our group's EFP2 calculations in GAMESS on 88 interactions in the S22 and S66 benchmark sets.^{2,3}
- 81 successfully processed, 7 left out due to time constraints
- Three files:
 - Calc_lmo** containing Localized Molecular Orbital (LMO) data
 - Calc_summary** containing interaction energies for each part
 - Sapt_benchmark** containing high-level reference data



References

- (1) Gordon, M. S.; Freitag, M. A.; Bandyopadhyay, P.; Jensen, J. H.; Kairys, V.; Stevens, W. J. **2001**, *105* (2), 293–307.
- (2) Jurecka, P.; Sponer, J.; Cerný, J.; Hobza, P. *Physical chemistry chemical physics : PCCP* **2006**, *8* (17), 1985–1993.
- (3) Řezáč, J.; Riley, K. E.; Hobza, P. *Journal of Chemical Theory and Computation* **2011**, *7* (8), 2427–2438.